MILITARY SPECIFICATION

PUMPS, CENTRIFUGAL, BOILER FEED, (MULTI-STAGE)

1. SCOPE

1.1 Scope. This specification covers horizontal and vertical, centrifugal, multi-stage boiler feed pumps, complete with drivers and controls.

1.2 Classification. Boiler feed pumps shall be of the following types, as specified (see 6.1):

Type I - Pump, centrifugal, split case, horizontal.
Type II - Pump, centrifugal, split case, vertical.
Type III - Pump, centrifugal, barrel type, horizontal.
Type IV - Pump, centrifugal, barrel type, vertical.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein;

SPECIFICATIONS

FEDERAL
FF-B-171 - Bearings, Ball, Annular (General Purpose).
FF-B-185 - Bearings, Roller, Cylindrical: and Bearings, Roller-, Self-Aligning.
FF-B-187 - Bearings, Roller, Tapered.
TT-P-28 - Paint, Aluminum, Heat Resisting (1200°F).
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.
GGG-E-950 - Extractor, Stuffing Box and Pump Packing (and Tamper), and Extractor," Lantern Gland.
GGG-P-781 - Puller, Mechanical, Puller Attachment, Mechanical, and Puller Set, Mechanical.

MILITARY
MIL-S-901 - Shock Tests, H. I. (High-Impact); Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-D-1000 - Drawings, Engineering and Associated Lists.
MIL-D-1000/2 - - Drawings, Engineering and Associated Lists.
MIL-C-2212 - Controllers, Electric Motor, A.C. or D.C. and Associated Switching Devices, Naval Shipboard.
MIL-P-15024 - Plates, Tags and Sands for Identification of Equipment.
MIL-P-15024/5 - - Plates, Identification.
MIL-M-15071 - Manuals, Technical: Equipments and Systems Content Requirements for.
MIL-E-15090 - Enamel, Equipment, Light-Gray (Formula No. 11).
MIL-P-15137 - Provisioning Technical Documentation for Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-P-15328 - Primer (Wash), Pretreatment, Blue (Formula No. 117-B for Metals).
MIL-P-16789 - Preservation, Packaging, Packing and Marking of Pumps General, and Associated Repair Parts.
MIL-M-17060 - Motors, 60 Hertz, Alternating Current, Integral Horsepower (Shipboard Usa).
MIL-L-17331 - Lubricating Oil, Steam Turbine (Noncorrosive).
MIL-M-17413 - Motors, Direct Current, Integral H.P. Naval Shipboard.
MIL-T-17523 - Turbine, Steam, General Auxiliary (Naval Shipboard Use).
MIL-P-17545 - Primer Coating, Alkyd-Red Lead Type, Formula No. 116 and Formula No. 116D.
MIL-B-17931 - Bearing, Ball, Annular, for Quiet Operation.
MIL-G-18709 - Grease, Ball and Roller Bearing.
MIL-G-18916 - Governors, Steam Driven Pump, Pressure Regulating, Naval Shipboard.

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MILITARY (Cont'd.)
MIL-S-22473 - Sealing, Locking, and Retaining Compounds; Single-Component.
MIL-G-24139 - Grease, Multipurpose Quiet Service.
MIL-I-45208 - Inspection System Requirements.

STANDARDS

MILITARY
MIL-STD-167 - Mechanical Vibration of Shipboard Equipment.
MIL-STD-271 - Nondestructive Testing Requirements for Metals.
MIL-STD-278 - Fabrication Welding and Inspection; and Casting Inspection and Repair for Machinery, Piping and Pressure Vessels in Ships of the United States Navy.
MIL-STD-438 - Schedule of Piping, Valves, Fittings, and Associated Piping Components for Submarine Service.
MIL-STD-740 " Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-777 - Schedule of Piping, Valves, Fittings and Associated Piping Components for Surface Ships.
MS16142 - Boss, Gasket Seal, Straight Thread Tube Fittings, Standard Dimensions for.
MS18229 - Plug for "O" Ring Gasket.

DRAWINGS
B-153 - Packings and Gaskets, Standard Application of.
B-214 - Root Connections for Attaching Piping.
810-1385850 - Piping, Gage for All Services.

PUBLICATIONS

MILITARY
NAVSHIPS 0900-001-7000 - Fabrication and Inspection of Brazed Piping Systems.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other ublications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL BUREAU OF STANDARDS
Handbook H28 - Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D. C., 20402.)

HYDRAULIC INSTITUTE
Test Code of Standards - Centrifugal Pump Section.

(Application for copies should be addressed to Hydraulic Institute, 122 E. 42nd Street, New York, New York 10017.)

(Technical society and technical association specifications and standards are generally lvailable for reference from libraries. They are also distributed among technical groups end using Federal agencies)

3. REQUIREMENTS

3.1 Qualification. The pumps furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at time set for opening of bids (see 4.2 and 6.5).

3.2 General design. The design and construction of all pumps shall comply with the space and weight requirements when specified (see 6.1). The pump unit shall be designed to perform under the list, pitch, roll and trim conditions as specified (see 6.1). Pumps shall be capable of operating continuously at any point over the entire flow range and the design conditions as specified (see 6.1).

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3.2.1 Reliability. The requirements of this specification are imposed exclusively to obtain equipment of utmost reliability for the intended service. The assurance of maximum reliability shall be the paramount controlling principle in the design. fabrication. assembly and testing of this l quipment. Users of and suppliers to this specification are urged to communicate to Naval Ship Engineering Center (NAVSEC) any findings related to the requirements or lack of requirements of this specification whereby improvement of equipment reliability can be achieved.

3.2.2 The equipment specified herein shall be operated, maintained and repaired on-board Navy ships. Attention is directed to the high equipment density of shipboard machinery spaces, to the fact that maintenance and repair must be made underway in heavy seas, that equipment must operate unattended, and that maintenance personnel may not be seasoned mechanics. The requirement for maximum reliability directly related to these shipboard environmental and service conditions and they shall be fully considered in the pump design. The aspects of "human engineering" shall also be fully explored and considered in the equipment design to minimize the possibility of failure through improper operation and maintenance and to preclude personnel safety hazards. In no ease shall parts be physically interchangeable or reversible unless such parts are also interchangeable functionally and from considerations of strength.

3.2.2.1 The equipment shall be designed for maximum repair accessibility, for ease of examination of wearing parts and for simplicity of disassembly and proper reassembly. Positioning and alignment of parts in assembly shall employ positive means such as shoulder, tongue and groove or other locating techniques whereby correct reassembly is repeatedly assured. In consonance with the concept of ease of maintenance and where not already specified, the pump designer shall consider all of the following features for incorporation in the design and technical documentation wherever such incorporation will significantly contribute to ease of maintenance without compromising performance reliability:

(a) Lifting lugs.
(b) Casing assembly guide pins.
(c) Alignment and positioning dowels.
(d) Jacking screws.
(e) Ease of access to couplings.
(f) Re-makeable piping connections.
(g) Piping out of way of maintenance access.
(h) Ease of packing, seal cage, mechanical seal, bearing and wearing ring replacement.
(i) Split casings, bearing housing, etc.
(j) Guards over rotating and moving components.
(k) Warning places to prevent casualties to equipment and personnel.
(l) Provisions for connecting instruments for performance evaluation.
(m) Attached instruments for monitoring performance.
(n) Visibility and access to attached instruments.
(o) Thoroughness of operating instructions.
(p) Thoroughness of preventive maintenance instructions.
(q) Explicit assembly and disassembly instructions.
(r) Exploded views of critical assemblies.
(s) Adequate system diagrams.

3.2.2.2 Each pump shall be designed to have a service life of 30 years of which 40 percent shall be actual operation. There shall be no limit on the number of starts during the life of the pump. It shall be assumed that during the life of the pump, parts subject to unavoidable wear and deterioration (with the exception of packing and rolling contact bearings) will be replaced at intervals not shorter than 12 years. The parts subject to wear, deterioration and normally requiring replacement at 12 year intervals during the s ervice life of the pump (with the exception of packing and rolling contact bearings) shall be designed for a life of 40,000 l atual operating hours, and they shall be identified in appropriate drawings and manuals. The requirements set forth above shall not be construed or interpreted as a warranty requirement nor otherwise affect the manufacturer's warranty.

3.2.3 Shock and vibration.

3.2.3.1 The boiler feed pump and driver with all appurtenances and controls shall be capable of passing a shook test in acordanoe with MIL-S-901 as specified in 4.3.3.2.

3.2.3.2 Shock mounts (resilient mounts for shook attenuation) shall not be used unless specified (see 6.1). When noise attenuation mounts are specified (see 6.1), the pump shall be designed to pass the specified shock test on the noise attenuation mounts.

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3.2.3.3 Pumps shall not be damaged or caused to malfunction either by internally excited vibrations, or by the environmental vibrations specified in MIL-STD-167. Frequencies shall be as specified (see 6.1).

3.2.4 Alignment.

3.2.4.1 The design of all pumps shall be such that alignment will not be disturbed or undue stresses set-up in any part by normal vibration, contraction and expansion of piping attached thereto in Service; The pump shall be designed to withstand loads imposed by attached piping as specified (see 6.1).

3.2.4.2 Alignment between separate components of a unit shall be maintained by means of keyways or other adequate means. All parts manufactured to the same drawings shall be interchangeable.

3.2.4.3 In no case shall a piece of equipment be rigidly supported from more than one plane.

3.2.5 Noise.

3.2.5.1 The design, construction, and workmanship, of the equipment shall be such that a minimum practical noise level will result under all operating conditions. Airborne noise, structureborne vibration and fluidborne noise requirements shall be as specified (see 6.1).

3.3 Detail requirements.

3.3,1 Performance characteristics.

3.3.1.1 The operating conditions and characteristics of the pumps shall be as specified (see 6.1).

3.3.1.2 The head-capacity curve of these pumps shall rise continuously from rated capacity to shut-off so that at constant rated speed the shut-off pressure shall be not less than 15 percent greater than at rated capacity.

3.3.1.3 At constant rated speed the pumps shall develop 120 percent rated capacity at a pressure not less than 80 percent of rated discharge, pressure unless otherwise specified (see 6.1).

3,3.1.4 All feed pumps shall be designed to operate in parallel with other identical pumps without surging, under all conditions of operation from shut-off to full capacity when operating under any of the following methods of control of the driving unit:

(a) Unit operating at constant speed under control of speed limiting governor.
(b) Unit operating at variable speed under control of pressure governor.
(c) Unit operating at constant or variable speed with electric motor.

3.3.1.5 The limiting rotative speed for these pumps for a given capacity, suction pressure and temperature shall be as specified (see 6.1).

3.3.1.6 The steam pressure drop through pump pressure regulating governors shall be considered when rating turbine driven pumps.

“3.3.1.7 Motor driven pumps shall be able to withstand reverse rotation for a period of one minute at all speeds up to design speed without damage.

3.3.2 Casings.

3.3.2.1 Casings shall be of the double volute, diffuser concentric or modified concentric casing design to minimize radial bearing loads.

3.3.2.2 Stationary (casing) wearing rings shall be fitted to all pumps.

3.3.2.3 Casings of horizontal and vertical pumps shall be provided with flanged vent connections. The vents shall be located to permit release of entrapped air prior to start-up to ensure complete submergence of the rotor in the process fluid. Drain connections shall be l ither flanged or threaded in accordance with MS16142 and MS18229 and shall permit complete drainage of the pump without disassembly of the pump. Vent and drain valves shall be supplied with the unit unless otherwise specified (see 6.1).

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3.3.2.4 The requirements specified in 3.3.2.4.1 through 3.3.2.4.5 shall apply to casings of type I and type II pumps.

3.3.2.4.1 The casings shall be divided into two parts so as to permit ready removal of the rotors and for bearing and gland replacement.

3.3.2.4.2 The design of casings shall be sufficiently rugged to withstand without fracture or appreciable distortion the strains to which they may be subjected.

3.3.2.4.3 Clearance shall be provided around bolt heads and nuts to permit the use of ordinary tools.

3.3.2.4,4 Fitted bolts or heavy dowel pins shall be provided to insure maintenance of alignment of casing halves in reassembly. Tapered dowel pins shall be corrosion-resistant, shall have a threaded end and nut design, and shall be secured against coming adrift under shock loading.

3.3.2.4.5 Forcing bolts shall be provided for breaking joints.

3.3.2.5 The outer casing of type III and type IV pumps shall be a solid barrel sufficiently rugged to withstand, without fracture *or* distortion, the strains to which it may be subjected.

3.3.3 Suction and discharge connections. The suction and discharge connections shall be flanged in accordance with MIL-STD-438 for submarines, MIL-STD-777 for surface ships, or as otherwise specified (see 6.1). Suction and discharge connection sizes, orientation? location and surface finish shall be as specified (see 6.1).

3.3.4 Impellers.

3.3.4.1 Outside surfaces of impellers shall be smooth finished.

3.3.4.2 Impellers shall be keyed on the shaft and securely held against lateral movement by locked nuts or other means approved by the design review agency.

3.3.4.3 Each impeller shall be dynamically balanced.

3.3.4.40 Impellers shall not be fitted with wearing rings. The impeller hub wearing " surfaces shall have material thickness to permit reducing the diameter of the impeller hubs by as mush as 0.050 inch to accommodate undersize easing wearing rings to restore design running clearances.

3.3.4.5 The radial clearance between impeller hub wearing surfaces and the casing wearing rings shall be not less than that specified in table I. All clearances to be measured on the radius; that is, the difference in diameter between wearing surfaces shall be not less than twice the clearance in table I.

Table I - Radial clearance

Rated capacity	Radial clearance, minimum
Gpm	Inch
249 and below	0.0060
250 to 499	.0075
500 and above	.0090

3.3.5 Pump shafts.

3.3.5.1 All shaft threading, except threading for rolling contact bearing locknuts, shall be counter to the direction of rotation. Shaft threading for rolling contact bearing locknuts may be clockwise *or* counterclockwise.

3.3.5.2 Bearing locknuts shall be provided with a positive means of locking.

3.3.5.3 Shafts shall be fitted with shaft sleeves in way of stuffing box, diaphragm bushings and water lubricated bearings. A flinger shall be so located that leakage which sight occur between the stuffing box shaft sleeve and the shaft, as well as from the stuffing box, is not allowed to reach pump or motor ball bearings.

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3.3.5.4 "O" ring seals shall be used between the shaft and the stuffing box shaft sleeve. Other positive methods of sealing may be used subject to the approval of the design review agency concerned.

3.3.5.5 Keys shall be used in the shaft to avoid relative rotation of rotor components, such as impellers, sleeves, spacer pieces and couplings.

3.3.5.6 Impellers, shafts and associated hardware in the rotating assembly shall not contain radially oriented set screws.

3.3.6 Rotating assembly.

3.3.6.1 All rotors shall be dynamically balanced with all rotating parts connected thereto; for common shaft assemblies or those using a rigid coupling, this requires dynamic balance with the rotating elements of the driving unit in place. However, rotating parts may be balanced individually provided that when assembled the unbalance shall not exceed the limits specified in MIL-STD-167. Where balancing is required as a maintenance procedure to maintain proper vibration or noise performance of the pump and driver unit in service, the pump shall be designed such that balancing may readily be performed by maintenance personnel.

3.3.6.2 The pump design shall be such and calculation shall be made to demonstrate that the maximum operating speed of the pump will not be greater than 70 percent of the first critical speed. If this proves to have an impractical effect on mechanical or -hydraulic design, the unit shall be designed for operation above 125 percent of the first critical speed. The second critical speed shall be not less than 30 percent above the maximum operating speed.

3.3.6.3 All pumps shall incorporate design features to achieve hydraulic balance to obtain a thrust bearing life of 10,000 hours or greater.

3.3.6.4 Barrel-type pumps shall be so designed that the assembled pump rotor (exclusive of the coupling, including all non-rotating parts which can be installed only by disassembling rotating parts) can be removed, replaced and stored as a complete assembly. This assembly shall be capable of passing through a hatch of the specified diameter (see 6.1).

3.3.7 Stuffing boxes and mechanical shaft seals.

3.3.7.1 All pumps except those on nuclear ships shall be provided with mechanical shaft seals. The gland of each seal shall be designed to incorporate two or more rings of packing for use in the event of a mechanical shaft seal failure. Seal internals shall be constructed of 304 or 316 stainless steel. The mechanical shaft seal shall have a tungsten carbide against carbon sealing surface.

3.3.7.2 The design shall be such that the pressure on the seal or stuffing boxes will not exceed pump suction pressure by more than 50 pounds per square inch (p.s.i.). Replaceable pressure breakdown sleeves and bushings shall be separately fitted to the shaft and casing fitted with packing. Leek-off connections shall be between the pressure breakdown device and the stuffing box. The Leak-off shall be led to the suction side of the pump with the point of entrance properly located to avoid flow disturbance to the first stage impeller. Gage confectioner shall be supplied as specified (see 6.1) and located in such a manner as to accurately determine the pressure on the stuffing boxes or seal. All stuffing boxes l xcept those with mechanical seals shall be fitted with throat bushings. For packed pumps all stuffing box glands shall be set up by nuts threaded on gland studs secured in casing. Sufficient space shall be provided between bearings and stuffing boxes to permit easy renewal of packing without disturbing bearings, and inspection end overhaul of. bearings. The gland design shall insure that gland leakage cannot reach the bearings or be thrown over bearing housings and driving unit. This shall be accomplished preferably by use of internally cored glands and by the use of shaft flingers. Ample drip pockets, spray shields and drains shall be provided. Glands shall be split in halves with the halves secured together in a manner to form a solid ring for setting up.

3.3.7.3 Space shall be provided between bearin s and stuffing boxes to permit easy renewal of packing inspection end overhaul of mechanical seals without disturbing bearings, end examination and overhaul of bearings.

3.3.7.4 Mechanical shaft seals shall be positioned on the shaft by means of stub or step sleeves. Mechanical shaft seals shall not be positioned by Use of set screws.

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3.3.7.5 For mechanical shaft seals, the design shall insure that positive liquid pressure is supplied to the seal faces under all conditions of operation and that there is adequate circulation of the liquid at the seal faces to minimize the deposit of foreign matter on the seal parts. The pump certification data shall contain certification by the seal manufacture that there will be adequate circulation of liquid and adequate lubrication at the seal faces when the seal is installed as shown on the drawing.

3.3.7.6 The design shall insure that any Leakage cannot reach the bearings or be thrown over bearing housings and driving unit. Drip pockets and drains shall be provided.

3.3.8 Bearings.

3.3.8.1 Radial bearings.

3.3.8.1.1 Radial bearings may be of the hydrodynamic or rolling contact type (see 3.3.9). If specifically approved by the design review agency hydrostatic bearings may be used (see 3.3.8.1.4).

. 3.3.8.1.2 Hydrodynamic radial bearings may be of the fixed or self-aligning type and shall be designed to operate under full fluid film (stable lubrication) operation over the entire range of the characteristic curve as defined by the operating conditions. Where the requirement for full fluid film operation cannot be met, a hydrostatic bearing may be recommended by the vendor (see 3.3.8.1.4).

3.3.8.1.3 Water lubricated hydrodynamic radial bearings shall not be provided with lubrication grooves unless specifically requested by the vendor and approved by the design review agency. If a hydrodynamic bearing design having axial grooves is recommended by the vendor, the actual direction of radial loadin must be established. The orientation of the bearing shall be such that the effective bearing arc will not be reduced to less than 120 degrees at any operating condition of the pump. The effective arc is defined herein as the continuous load carrying surface which is symmetrical with respect to the load vector. Release for manufacture shall be subject to the verification of the applied load direction either by a method approved by the design review agency or by l xperimental data approved by the design review agency l nd derived from tests of manufacturer's own similar designs.

3.3.8.1.4 Where the requirements.for full fluid film operation (see 3.3.8.1.2) of a hydrodynamic bearing cannot be met, as an alternative the vendor may recommend a hydrostatic bearing for approval of the design review agency. The hydrostatic bearing shall operate without metal-to-metal contact over the entire operating range.

3.3.8.1.5 The hydrodynamic and hydrostatic bearing surface and the journal shall be finished to 32 roughness height rating (RHR) or better.

3.3.8.1.6 A radial bearing shall consist of a housing, shell and liner. The housing shall be of a split design consisting of a cap bolted to a base and shall contain the bearing shell and liner. The liner may be l ither cast in the shell or may be a removable insert.

3.3.8,1.7 Each bearing housing shall be supported by a bracket that is either cast integral with the casing, securely bolted to the casing on a machined shoulder, or held in alignment with the casing by at least two heavy dowels and securely bolted thereto. The use of bolts alone for scouring brackets is prohibited.

3.3.8.1.8 Hydrodynamic radial bearing shells shall he split along the axis and so arranged as to permit renewal or refitting without removal of pump rotor from the casing. They shall be neatly fitted in their seatings to prevent axial play. Shells and their seatings shall be finish machined.

3.3.8.3.9 Internal water lubricated hydrodynamic radial bearings, for vertical pumps only, shall be of the one piece sleeve type.

3.3.8.2 Thrust bearings.

3.3.8.2.1 A thrust bearing shall be installed on l ach pump to counteract any unbalanced hydraulic or mechanical thrust in either direction. In this connection, consideration shall be given to the fact that rolling, itching, or list of a ship at sea may introduce thrust loads in a unit which is in hydraulic balance.

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3.3.8.2.2 Thrust bearings may be of the combined radial and thrust rolling contact type (see 3.3.9) or pivoted segmental type.

3.3.8.2.3 When pivoted segmental type thrust bearings are used, the design shall be such as to permit renewal or refitting of shoes without removal of shafts.

3.3.8.2.4 When a rolling contact thrust bearing is used, the design shall be such as to permit replacement of the bearing without removing the pump rotor from the casing.

3.3.8.2.5 For vertical flexible coupled pumps, thrust bearings shall be installed at the top, external of the pump, between the coupling and the stuffing box. For rigidly coupled pumps, the thrust bearing shall be located in the driver.

3.3.9 Rolling contact bearings.

3.3.9.1 Except as otherwise approved by the design review agency, rolling contact bearings shall be of the ball type in accordance with FF-B-171. Roller bearings may be recommended if use of ball bearings is impractical. If approved, roller bearings shall be in accordance with FF-B-185 or FF-B-187. Rolling contact bearings shall be selected to result in a minimum B-10 life of 10,000 hours calculated in accordance with FF-B-171, FF-B-185 or FF-B-187.

3.3.9.2 Rolling contact bearings specifically selected for quiet operation in accordance with MIL-B-17931 shall be used for pumps for submarine applications and for surface ships where required to meet special noise requirements.

3.3.9.3 Rolling contact bearings having a DN value in excess of 200,000 shall be oil Lubricated, unless the vendor can demonstrate to the satisfaction of the design review agency that the specific grease lubricated bearing application has been proven in previous Navy shipboard service. Rolling contact bearings having a ON value in excess of 270,000, shall not be used unless specifically approved by the design review agency.

Where D - Diameter of bearing bore in millimeters
N = RPM

3.3.9.4 Rolling contact bearing lubrication method shall prevent bearing failure resulting from overgreasing. Rolling contact bearing temperature shall not exceed **180°F.** (as measured on the bearing outer race) with specified normal pumped fluid temperature and specified maximum ambient air temperature outside the pump envelope. Straight holes with plugs shall be provided in the bearing housing to permit insertion of a thermocouple probe to contact the outer race diameter of each rolling contact bearing. Unless otherwise specified (see 6.1), externally supplied cooling liquid or externally induced forced draft shall not be required for satisfactory bearing temperature.

3.3.10 Couplings.

3.3.10.1 Boiler feed pumps shall be of 4-bearing construction with a flexible coupling between pump and driving unit. The use of other than 4-bearing construction required design approval and will be considered only when the vendor can substantiate that satisfactory alignment and bearing loads will be maintained. Rigidly coupled units may be used when specified (see 6.1).

3.3.10.2 Each flexible coupled unit shall have an all metal spacer type flexible coupling as specified (see 6.1) connecting the pump rotor to the driving rotor. Drawings shall indicate by note the maximum allowable backlash before replacement of the coupling is necessary.

3.3.10.3 All flexible coupling hubs shall be keyed to the shafts and secured by lock nuts. The hubs shall be fitted on a taper with keys.

3.3.10.4 All rigid couplings shall have fitted bolts and the coupling flanges shall be marked at assembly to insure proper reassembly after overhaul of the unit.

3.3.10.5 Coupling guards shall be provided for each horizontal flexible or rigid coupled unit. Guards shall be designed to permit ready access to the coupling for lubrication, examination and maintenance.

3.3.11 Mounting bedplates.

3.3.11.1 Each horizontal pump and its driving unit complete with all appurtenances shall be mounted on a common bedplate.

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3.3.11.2 Bedplates of horizontal flexible coupled units shall be sufficiently rigid to permit handling, shipment and installation of the unit onboard ship to minimize misalignment of the assembled unit. The driver shall be doweled and the unit realigned by the installing activity after final installation. The manufacturer shall furnish a warning plate indicating the above.

3.3.11.3 All bearing and seating surfaces of bedplates and mounting plates shall be finish machined.

3.3.11.4 Each component part of an assembled horizontal flexible coupled unit supported directly by the bedplate shall be doweled thereto to facilitate reassembly and maintenance of alignment, except that final doweling of the driver shall be performed by the installing activity after final installation.

3.3.11.5 Vertical pumps shall be mounted as close to the center of gravity as practical.

3.3.12 Lubrication.

3.3.12.1 A simple, positive, and self-contained forced feed lubrication system shall be provided on each turbine-driven pump assembly in accordance with MIL-T-17523, oil temperatures in the bearings shall not exceed **180°F.** (measured by thermocouples) under any condition of operation.

3.3.12.2 Grease lubricant shall be in accordance with MIL-G-18709 or for quiet bearings, MIL-G-24139. Oil lubricants shall be in accordance with MIL-L-17331. The design shall be such that any qualified grease under MIL-G-18709 or oil under MIL-L-L7331 can be used interchangeably with any other qualified product to the same specification.. Bearings shall be capable of being relubricated without removing the bearings and shall not require frequent relubrication if the pump is not operated over an extended period of time.

3.3.12.3 Oil-lubricated bearings shall be separate from and independent of the stuffing boxes, adequately protected from gland leakage. Means shall be provided to prevent the escape of oil around the shaft. All baffles and wipers shall be readily renewable.

3.3.12.4 Bearing housings shall be so designed that the bearing will be adequately lubricated immediately when the shaft starts to turn. this shall be accomplished by properly locating oil supply and drain pipes, and adequate sealing of the bearing housing so that regardless of the Length of time the unit is idle, an adequate supply of oil will remain in the bearing housing.

3.3.13 Eyebolts, lugs, holes and other means approved by the design review agency, shall be provided to permit attachment of lifting gear for lifting the assembled pump, driver and attached accessories as a complete unit. Means approved by the design review agency shall be provided for the handling of parts and components weighing 150 pounds and over which cannot be handled manually during unit overhauls and preventive maintenance inspections.

3.4 Materials. The materials of the pump shall conform to the materials specified in 6.3. However, this specification is not intended to be restrictive provided proposed alternate materials will give equal or better service than the materials specified. Proposed alternative materials shall be subject to approval by the design review agency. Components of the pump for which specific materials are not specified shall be of material best suited for the service intended. Materials which can be sensitized and are subjected to heat treatment in the sensitization range during fabrication shall be capable of passing the intergranular corrosion tests of the base material specifications. Particular attention should be given to avoiding sensitization of materials during hard facing, stress relieving, or repair welding.

3.4.1 All valves, flanges, and bolting shall conform to MIL-STD-438 for submarines and MIL-STD-777 for surface ships and l s specified (see 6.1) except tapered pipe threads are acceptable for use in grease fittings located on non-moving parts. Fittings not covered by these Specifications for a pplication where ship connections are not involved shall be approved by the design review agency.

3.4.1.1 Where practical, the use of silver brazed joints should be avoided. However, where unavoidable, silver brazed joints shall conform to the fabrication and inspection raquirements of NAVSHIPS 0900-001-7000.

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3.4.2 Welding shall be in accordance with MIL-STD-278.

3.4.3 Threaded parts such as bolts, studs, and nuts shall conform *to* Handbook H-28. The use of tap bolts or cap *screws is* prohibited unless specifically approved by the design review agency, where the use of through bolts or studs is impracticable. The setting end of studs shall be class 5 fit or class 3 fit with locking resin in accordance with MIL-S-22473. The nut end shall be class 3 fit.

3.4.4 All gaskets shall be in accordance with Drawing B-153. Gasket compression and tolerances shall not effect baaring, bushing, or wearing ring fits to an extent that will adversely affect reliability and performance. Internal gasketing of type III and IV pumps shall be completely encased. Spiral wound gaskets shall be used only where use of a sheet gasket or O-ring is impracticable. Internal spiral wound gaskets shall be completely encased to prevent the gasket from unwinding and entering the fluid system.

3.4.5 Root connations for attaching pipes shall be in accordance with Drawing B-214.

3.4.6 Pipe connections shall be of the flanged, union, or welded type as applicable. Should this not be feasible, a straight thread fitting of a type approved by the design review agency, may be used.

3.4.7 Pressure gages shall be provided with the unit, unless otherwise specified (see 6.1), and mounted in accordance with Drawing 810-1385850. Gages shall be positioned for accessibility for reading and maintenance. Gages shall be secured to parent equipment in such a manner as to prevent vibration, breakage and disconnection.

3.4.8 Locking devices. All internal threaded fasteners and threaded machine **elements shall be secured by** locking devices where continuous operation under the condition *of* shock, vibration, and temperature specified herein depend on maintaining tight connections of parts. Sat screws shall have locking devices. Locking devices shall be subject to approval by the design review agency.

3.5 Identification plates. Identification plates shall be furnished on each pump and shall be type A or B in accordance with MIL-P-15024 and MIL-P-15024/5, except that identification plates of plastic *or* aluminum are not acceptable.

3.5.1 Identification plates shall be secured to equipment with corrosion-resistant metallic screws or rivets.

3.5.2 Pump identification plates shall contain data as follows:

(a) Manufacturer's name.
(b) Manufacturer's model or type and size.
(c) Service application.
(d) Manufacturer's pump serial number.
(e) Salient design characteristics.
    (1) Capacity in gallons per minute (g.p.m.).
    (2) Total head in p.s.i.
    (3) Speed *of* shaft in revolutions per minute (r.p.m.)
    (4) Brake horsepower.
    (5) Test pressure.
    (6) Special data vital to the unit:
        a. Suction pressure.
        b. Submergence.
        c. Impeller diameter(s).
        d. Contract number (and item number for multiple unit orders).
        e. Federal stock number.
        f. NAVSHIPS technical manual number.
        g. Section for Defense Contract Administration Service (DCAS) stamp.

3.6 Painting.

3.6.1 For pumps operating at a temperature of **300°F** or less, all external unmachined Surfaces of ferrous metal parts except corrosion-resisting steel shall be thoroughly eleaned and coated with paint in accordance with the following specifications:

(a) One coat of pretreatment primer in accordance with MIL-P-15328.
(b) One coat of zinc chromate primer in accordance with TT-P-645, or one coat of red lead primer in accordance with MIL-P-17545.
(c) One finish coat of light gray equipment enamel in accordance with MIL-E-15090 except where thermal insulation is installed.

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3.6.2 For pumps operating at a temperature over 300°F., whether to be insulated or not, all external unmachined surfaces of ungalvanized ferrous metal parts shall be thoroughly cleaned and coated with two coats of heat-resisting paint in accordance with TT-P-28.

3.7 Pump drivers and accessories.

3.7.1 Turbines. All turbines for driving boiler feed pumps shall be in accordance with the requirements of MIL-T-17523 and as specified (see 6.1). Turbines shall be equipped with a low feed pump suction pressure trip to protect the feed pump in the event of loss of suction, unless such a device is incorporated in the pressure governor. The trip shall be designed to cause loss of turbine governor oil pressure whenever the feed Pump suction pressure drops to trip set-point. The low suction pressure trip shall be equipped with a manual reset. The trip shall be designed to have an adjustable set-point over the pressure range specified (see 6.1).

3.7.2 Pressure regulating governors. Pressure regulating governors for turbine driven boiler feed pumps shall be in accordance with MIL-G-18916 and of the type and characteristics as specified (see 6.1).

3.7.3 Recirculation. The pump will be installed with a recirculation line leading from the pump discharge back to the system deaerator, whereby a minimum through-flow is assured to prevent pump damage. The recirculation line including the flow limiting pressure breakdown device will be furnished by the installer. However, the pump supplier shall determine and indicate on appropriate installation drawings the amount of through-flow necessary to prevent pump damage when the pump is operating continuously at shut-off at rated design speed.

3.7.4 Motors. All motors for driving boiler feed pumps shall be in accordance with MIL-H-17060 for alternating current (a.c.), or MIL-H-17413 for direct current (d.c.) motors as specified (see 6.1). The horsepower rating of each motor shall be not less than the maximum brake-horsepower of the driven pump under any condition from shut-off to free delivery.

3.7.5 Controllers. All controllers furnished shall be in accordance with MIL-C-2212 for a.c. or d.c. controllers as specified (see 6.1). When specified, motor driven pumps shall have a low suction pressure trip designed to de-energize the motor at the pressure set-point specified (see 6.1).

3.8 Repair parts and tools.

3.8.1 Onboard. Repair parts shall be furnished and processed in accordance with MIL-P-15137 (see 6.2) and as specified hereinafter.

3.8.2 A set of repair parts (onboard) shall be furnished in accordance with method B (paragraph 3.3.2) of MIL-P-15137 (see 6.2) and for guidance in preparing provisional technical documentation shall consist of the parts specified in table II, and special tools and wrenches required for servicing of the pump.

Table II - Onboard repair parts.

Name of part	Quantity per set
Coupling bushings, pins and other wearing parts of complete couplings Packing, mechanical seal	100 percent complete replacement for all installed units per ship
Sleeve bearings Pivoted segmental bearings Undersize casing wearing rings Diaphragm bushings Lubricating oil pumps complete with drive shafting and gears Oil pump gears or rotors Stuffing box throat bushings Pressure breakdown drum bushings Couplings gear type	One to 4 units per ship - 100 percent replacement for one pump 5 to 8 units per ship - 100 percent replacement for two pumps 9 to 12 units per ship - 100 percent replacement for three pumps
Rotors, completely assembled, including shaft, impellers, casing wearing rings, casing diaphragms and bushings, drums, sleeves, keys, nuts, flinger, and complete coupling Diffusion vanes	100 percent replacement for one pump for all ships except submarines

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3.8.2.1 The pump rotor assembly shall be completely assembled ready for installation in the unit for which intended and shall include any non-rotating parts which can be installed only by removing some rotating part or parts of the pump rotor.

3.8.2.2 For pumps with shafts common to both pump and driving turbine, the repair turbine rotor and removable parts in way of turbine, including governor parts, shall be packed separately.

3.8.2.3 The quantities of individual repair parts shall not be reduced because of duplication of parts furnished to make the rotor assembly except that standard casing wearing rings shall not be required.

3.8.2.4 Casing wearing rings (see table II) shall be machined undersize by at least 0.030 inch on the diameter to permit machining and restoration of design running clearances between the impeller hubs and the casing wearing rings.

3.8.2.5 Type III pump. One rotor cradle shall be furnished for each machinery compartment in which the pumps are installed, as specified (see 6.1).

3.8.2.6 Tools. Special tools required for the maintenance and repair of the pump units shall be furnished. The number of sets of special tools shall be as specified (see 6.1). Special wrenches should be of forged steel with hardened jaws. Special tools are defined as those tools not listed in the Federal Supply Catalog. (Copies of this catalog may be consulted in the office of the DCAS.)

3.8.2.7 Tools which are available in the Federal Supply Catalog, such as common wrenches and standard pullers and e xtractors, will not be required: Standard complete or limited sets of pullers or extractors or parts of puller sets in accordance with GGG-P-781 and GGG-E-950 shall be identified as to type and use in notes on the outline or section assembly drawings. Special tools shall be detailed and included in the list of material.

3-.8.2.8 Each box containing onboard repair parts shall contain a list entitled "List of onboard repair parts". The list shall be in a format for use on outline drawings and in the instruction books. It shall be not less than nominal 8-1/2 by 11 inches in size.

3.8.2.8.1 The format of the list shall include a heading and columns of data for the items listed. The heading shall include titles and applicable entries as follows:

Reproduced from drawing number _____
Number of ships _____
Application _____
Contract number _____
Manufacture r _____
Quantities are for _____ units per ship
The columns shall include:
- Piece number _____
- Name of part _____
- Quantity _____
- Drawing number _____
- Federal stock numbe _____

Additional columns may be used as applicable.

3.8.2.8.2 The list shaLl be so treated as to be resistant to oil, water and fading.

3.8.3 The repair parts requirements for the drivers and accessories shall be in accordance with the related equipment specification.

3.8.4 Stock. The selection, stock numbering and quantities of repair parts for stock shall be determined and processed in accordance with MIL-P-15137 (see 6.2).

3.8.4.1 The cognizant supply demand control point for repair parts for stock is Ships Parts Control Center, Mechanicsburg, Pennsylvania, unless otherwise specified in the contract or order,

3.9 Drawings. Drawings shall conform to the requirements of categories A, G, H, I and J, form 2 of MIL-D-1000, and types II and III of MIL-D-1000/2, as specified (see 6.1).

3.9.1 Drawing content.

3.9.1.1 General requirements.

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3.9.1 .1.1 "Manufacturer's Use only" notes. Information intended for manufacturer's use only shall be so designated. Conversely on a commercial drawing with wide usage it is permissible to designate portions thereon, "For Navy use only".

3.9.1.1.2 Dimensional tolerance. Unless otherwise indicated by note, legend, or key, all tolerances shown on a drawing must be complied with during both manufacture and maintenance repair.

3.9.1.2 Design drawings. Design drawings shall consist of the drawings described in **3.9.1.2.1 through 3.9.1.2.4.** Design drawings shall be submitted to the design review agency for review except where submittal for approval is required (see 6.1.1).

3.9.1.2.1 Sectional assembl drawin s. Sectional assembly drawings shall include a sectional assembly with references to the list of material for identification of individual parts. Side or partial views may be required in order to show parts not otherwise shown in the main section view. All running clearances shall be shown and shall be dimensioned as radial clearances. (The radial clearance equals one half the diametral clearance.) Tightening torques and thread Lubrication requirements for threaded fasteners shall be shown on the assembly drawing.

3.9.1.2.2 List of material. The list of material shall include every part required in the pump a ssembly, including those parts not required to be detailed. The list of material shall include an indication of l ach part required by this specification to be furnished as an onboard repair part (this is not a repair parts list). The list of material shall include the manufacturer's drawing number and service part number for all parts for which detail drawings are required (see 3.9.1.2.3) as well as quantity required, material type, and material specification for all parts (see 3.9.2). Parts shall be named to indicate the function they serve. Fasteners and other parts for which detail drawings are not required shall be identified as to size and type. The list of material may be shown on the sectional assembly drawing.

3.9.1.2.3 Detail drawings. Detail drawings shall be furnished of l ll parts and sub-assemblies necessary for evaluation of the equipment, and of all parts necessary for maintenance, and overhaul of the pumps. Details of these parts shall be so complete as to permit emergency manufacture by a Naval shipyard without assistance from the original manufacturer. Details of pump casings shall be furnished, but the dimensioned development of cored hydraulic passages need not be shown. Sub-assemblies whose parts can not be procured or serviced individually, should be shown as a single part and so indicated. Multi-detail drawings are preferred, but monodetail drawings may be used.

3.9.1.2.4 Outline drawin s and certification data. Outline drawings and certification data shall be furnished as supplemental drawings to all sectional assemblies and details. Separate outline drawings, drawing lists and certification data shall be furnished under l ach contract or order unless the complete equipments covered by the outline drawing and the reference drawings are in fact identical in all respects.

t 3.9.1.2.4.1 The outline drawings in addition to the certification data required by MIL-D-1000/2 shall contain the following;

(a) Dimensional outline assembly drawing of the pump with its prime mover, bedplate and attached auxiliaries.
(b) Complete performance data of pump, prime mover, and attached auxiliaries, if l pplicable.
(c) Complete equipment performance curves, based on actual tests (the original submissions may show design performance curves versus test curves).
(d) Weights of major subassemblies over SO pounds, e.g. rotor, upper half-casing, lubricating oil cooler, and the weight of the complete unit, including driver, dry and in operating condition.
(e) Shipbuilder's connections showing size, type and dimensions of flanges.
(f) Center of gravity of the pump component and of the complete assembly.
(g) Radii of gyration of complete assembly about l aeh of the three principal axes.
(h) Identification of system in which installed.
(i) Critical speeds.
(j) Maximum thrust loads.
(k) The endurance test, high impact shock tests and non-destructive tests (if performed) and NAVSEC or design review l gency letters l pproving these reports or l xtension of any tests.
(i) Overall dimensions of the complete unit including locating dimensions of all fittings and connections, the space required for removal and replacement of parts for maintenance, end mounting information.

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3.9.1.3 Onboard repair parts list. A list of onboard repair parts shall be furnished with outline drawings. This List shall be prepared on Provisioning List forms in accordance with MIL-P-15137 (see 6.2). This list shall be furnished whether or not the parts are required in the pump contract or order. This list shall not be modified to indicate parts or quantities of parts furnished under supplementary or separate contracts or orders. The list is intended as a record of recommendations of the manufacturer and purchaser at the time of equipment drawing review. The list of onboard repair parts actually purchased shall be prepared as required by MIL-P-15137 (see 6.2).

3.9.1.4 Basic design drawings. Basic design drawings covering pumps under this specification are intended for submittal to NAVSEC prior to and independent of invitations for bids and as such they are different from design drawings as described in 3.9.1.2. Each drawing may cover a range of sizes of pumps so long as they are of the same basic design. A basic design drawing may indicate a variation in materials from that called for in 6.3, and if approved by NAVSEC that variation in material shall take precedence. over the corresponding material in 6.3. All such material variations shall be specifically pointed out in the transmittal letter.

3.9.1.4.1 Contents of basic desi n drawin s. These drawings shall consist of small scale (not necessarily to scale) plan and elevation views showing over-all dimensions. If the drawings cover a family of pump sizes all of the same design, the dimensions may be tabulated. The range of capacities, pressure and speeds shall be shown. The main part of this drawing shall be an undimensioned sectional assembly with complete list of material which shall include every part required in the pump assembly, If materials of some parts may change with various pump characteristics or with differing fluids to be pumped, a table shall be included to show these variations. Alternate subassembly arrangements of different parts may be included as desired.

3.9.1.4.2 Detail drawings of individual parts are not required in connection with basic design drawings, but may be included if desired by the manufacturer for clarification, or to obtain advance approval of NAVSEC.

3.9.2 Material identification.

3.9.2.1 Preferred material reference. Where materials of identical or equal quality can be identified by more than one specification or standard, the drawings need reference only one such specification or standard. In selecting the specification or standard to be referenced the following is the order of preference:

(a) Federal specification or standard.
(b) Military specification or standard.
(c) Industry and Technical Society specification or standard.
(d) Manufacturer's specification or standard.

3.9.2.2 Material substitutions. Where materials other than covered by (a), (b), and (c) of 3.9.2.1 are referenced and approved by the design review agency, the drawings shall show the complete chemical and physical properties of the approved material. In addition, the drawings shall identify the material in terms of the nearest Federal or Military specification, in order to enable Naval repair facilities to make emergency repairs as necessary for Navy material stocks.

3.9.3 Drawing identification. Prime vendors who purchase items from subcontractors shall use the subcontractor's drawing number as the single reference identification in all cases where the part(s) delineated thereon are produced by the subcontractor. The prime vendor shall not add his drawing number to the drawing except au an unofficial reference outside the drawing border or margin.

3.9.3.1 Prime vendors who purchase semi-finished parts from subcontractors for final production, test or selection phases in their own shops, have the option of using as the single drawing identification either their own title block and drawing number or the title block and drawing number of the subcontractor, but not both.

3.9.4 Drawings of drivers and associated equipment shall be in accordance with the specifications covering those equipments.

3.10 Technical manuals.

3.10.1 Unless otherwise specified in the contract or order, technical manuals shall be in accordance with type I of MIL-M-15071. A separate manual shall be furnished for each different pump unit l xcept in special cases when manuals covering more than one pump may be

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approved. Preliminary manuals shall be submitted to the design review activity for review (except where submittal for approval-is required (see 6.1.1)) and shall include all proposed sections complete, Unless otherwise specified (see 6.1), manuals shall include pump, driver, controls and all appurtenances and all applicable drawings. Performance curves for final manuals may be furnished as insert pages after delivery of hardware.

3.10.2 Manuals covering pumps driven by turbines or engines shall include drawings and full operation and maintenance instructions covering the turbines, engines, governors and any other applicable accessories.

3.10.3 Manuals covering pumps driven by conventional electric rotors with conventional controllers need not include operation and maintenance instructions covering the motors and controllers, except that complete motor bearing lubrication and replacement instructions shall be included. They shall, however, include master drawings and certification data covering the motors and controllers. Complete operation and maintenance instructions covering electrical equipment shall be included when required (see 6.1).

3.10.4 Each manual shall include not less than the following illustrations covering the pump and driver:

(a) Sectional assembly drawing.
(b) Outline drawing.
(c) Complete list of material corresponding to the sectional assembly drawing.
(d) Certification data.
(e) A minimum of two photographs of the complete unit, pump and driver, taken 180 degrees apart on a horizontal plane. Where several sizes of identical design units l re being supplied, photographs of one design unit only need be supplied.

3.10.5 The alignment procedure instructions in the manual shall describe in complete datail the means by which the required alignment clearance at sleeve bearings and wearing rings is to be established. The manual shall prescribe the maximum permissible measured eccentricities which can be tolerated in the aforementioned area with the pump in the as-assembled condition. Exceptions to any pert of the aforementioned requirements in any specific area may be had only by providing for the design review agency approval a written technical justification for the said l xception. The manual shall specify the maximum allowable clearances for wearing rings and sleeve bearings before replacement of the part(s). In addition the manuals shall contain:

(a) Lubrication schedule including lubricant and procedure.
(b) Detailed mechanical seal installation instructions. The minimum allowable leakage rate shall be specified.
(o) Threaded fastener torque values and sequence required to maintain proper seal and alignment.
(d) In-field dynamic balancing instructions where provisions for balancing in field are required to maintain noise and vibration characteristics.

3.10.6 The quantity and distribution of manuals shall be as specified (see 6.1).

3.11 Desi n data and calculations. Design data and calculations when required (see 6.1). shall be submitted approval of the design review agency for each pump type, design and size. Submittal shall be not later than submittal of design drawings for review. It is intended that the design data and calculations shall be in sufficient detail to describe the design, physical appearance and internal assembly to permit release for manufacture. References listing the author, publication, volume or test for all assumptions made in the calculations and for all formulas or methods of computation not readily identifiable shall be cited.

3.11.1 Design performance curves. The following curves shall be provided:

" (a) Head versus flow.
(b) Hydraulic efficiency versus flow.
(c) Hat positive suction head required versus flow.
(d) Brake horsepower versus flow.

3.11.2 Design features. A brief discussion of the following design features shall be provided:

(a) Number of stages and reasons for selection.
(b) Method of shaft l caling.

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(c) Type of coupling including previous service experience.
(d) Those areas in which system design conditions, as furnished by the procuring activity, adversely affect pump reliability or result in abnormal design features.
(e) Steps required for disassembly and reassembly (including alignment) to perform routine maintenance.
(f) Any other design features which the manufacturer considers of significant interest to describe the proposed unit.

3.11.3 Use of materials other than as specified herein shall be justified. It is not intended that materials which conform to and are used as specified herein shall require justification.

3.11.4 Calculations shall be submitted to cover the following:

(a) Shafting analysis of pump and driver assembly including calculation of critical speed. If calculations are done by computer, the method of analysis shall be described and only the results shall be submitted with identification of corresponding pump parameters used in the calculations. The following shall form the basis of the critical speed calculation:
(1) The effects of all major changes in shaft cross section moment of inertia shall be included.
(2) No support for the shaft shall be assumed other than at the designated bearings.
(3) All masses in the rotating system, including the effect of entrained fluids, shall be lumped at discrete points along the shaft. .
(4) No support for any shafting shall be assumed to be provided by additional shafting when two sections of shafting are separated by a flexible coupling, nor should any flexibility in bending or shear be assumed for a rigid coupling.
(5) For pumps which operate under variable speed conditions, the operating speed shall be construed to be the maximum operating speed.
(b) Loads imposed on the shaft and bearings. The calculations shall include the following:
(1) Axial and radial loads and load directions. Loading should be calculated for the design point and also for the highest loaded conditions over the operating range. These loads shall include all operating loads together with those that might be induced by thermal expansion, by bearing preload and by build-up of tolerances in the completed assembly. The radial loads at the impeller shall be calculated by the following:

$$P = \frac{KHD_2B_2}{2.31}$$

Where:

P = the radial load in pounds.
H = head in feet.
D_2 = impeller outside diameter, inches.
B_2 = impeller overall width including shroud, inches.
K = a constant which varies with capacity.

Unless data, supported by tests, justifies the selection of a lower value, K shall be in accordance with the following. Lower K values may be used only if specifically approved by the design review agency.

Casing type	At shut off	At rated flow	At 140 percent rated flow
Single volute	0.00024Ns for Ns less than 1500	0.10	0.45
	0.36 for Ns of 1500 or greater		
Double volute		0.05	0.10
Concentric	0.07	0.36	0.50
Modified concentric	0.15	0.15	0.15
Vaned diffuser in volute	0.15	0.07	0.15
Vaned diffuser in concentric	0.07	0.03	0.15

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Where:

Ns = specific speed at the best efficiency point.

(2) Effects of thrust reversal where conditions which might cause thrust reversal exist.

(3) Effects of roll, pitch, list and trim of the ship (including gyroscopic effects) shall be considered in the bearing design.

(4) Bearing clearance ratios, L/D ratio and characteristic numbers, lubrication flow and temperature rise (water lubricated bearings).

(S) Minimum film thickness required and calculated under maximum radial and axial load conditions over the range of specified pump operating conditions (water lubricated bearings).

(c) Analysis of driver bearings for close-coupled or rigidly coupled units.

(d) Calculation of the worst case cumulative effect of dimensions and tolerances for parts and locations. These calculations shall demonstrate:

(1) There will be no interference between rotating and stationary parts as assembled.

(2) There will be no interference between rotating and stationary parts, and hence no reduction in life of wearing parts, when these worst case cumulative effects are combined with operational environmental, and thermal loading.

3.12 Design record. A design record when required (see 6.1) shall be submitted to the design review agency. This record shall be completed except as noted in 3.12.2, and forwarded prior to unit delivery or at a time agreed to by the design review agency concerned.

3.12.1 The design record shall contain the following:

(a) The design data and calculations which were-submitted for manufacturing release, except as modified to resolve comments at time of manufacturing release.

(b) Revised calculations in those areas *of* design where the original calculations must be modified as a result of an approved design change.

(c) Approved design l valuation test reports, including resolution of comments in the approval action and any special testing which may be performed to verify new design features.

3.12.2 Changes in pump design approved subsequent to issue of the design record, or approved deviations in the design evaluation test unit from the design described in the design record, shall be submitted as an addendum to the design record. The addendum shall include applicable revised drawings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for ins ction. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The inspection requirements shall consist of the following:

(a) Completed part examination (see 4.1.2),

(b) Qualification (see 4.2).

(c) Tests (see 4.3).

(1) Shop tests. To be conducted on each pump supplied under the contract or order (see 4.3.1).

(2) Performance l valuation tests. To be conducted on one pump *of* each type, design and size on each contract or order (ace 4.3.2).

(3) Design evaluation tests. To be conducted on one pump of each type, design and size (see 4.3.3).

" The inspection system which the supplier is required to maintain, as provided in inspection clause of the contract or order, shall be in accordance with MIL-I-45208 (see 6.1 and 6.2).

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4.1.2.1 The inspection system shall include a completed part examination procedure whereby each functionally and dimensionally critical part and assembly, including the final product, is l xamined on completion in accordance with a completed part examination check list accompanying each part or assembly. The examination shall be performed by inspection personnel not engaged in the fabrication and assembly of the part. The check list for each part shall include accurate part identification and shall list the specific attributes which determine the parts acceptability. The check list shall be filled in during the time of examination. The check list shall be arranged in tabular form, calling forth separately the following examination assignments, each accompanied by the specific attributes of acceptability which shall be checked or measured and the findings recorded:

(a) Dimensions. They shall be as indicated on the approved design drawings. Rotating parts shall be examined for balance and maximum permissible run out as indicated on the approved drawings.
(b) Materials. Visual examination for compliance with approved design drawings. Any appearance of non-conformance shall be further verified by appropriate tests.
(c) Cleanliness. Examination for cleanliness shall apply equally to parts examination prior to assembly as well as subassemblies and the completed product.
(d) Missing parts: This examination applies primarily to the completed product . prior to shipment.
(e) Assembly. During assembly positioning tolerances and wear ring/impeller running clearances shall be verified.
(f) Alignment. The l quipment and its subassemblies shall be examined to insure that the alignment is as specified.
(g) Adjustment. The adjustment of safety, control and monitoring devices shall be verified and the settings recorded. Settings which should not be tampered with, once adjusted, shall be tagged with an appropriate precautionary warning.
(h) Preservation. The preservative used, the method of application, and recommended procedure for removal of preservative prior to placing the unit in service.

4.1.2.2 A copy of the completed part examination check list for the completed pump identified by manufacturers' serial number, shall be furnished the procuring agency in duplicate for each pump. On these copies the suppliers' quality control manager shall certify that the inspection requirements specified herein have been fulfilled and that the unit is in accordance with the approved drawings and the applicable specifications.

4.1.3 Variations.

4.1.3.1 Equipment variations. Any equipment, component or part which, during production or prior **to delivery,** deviates in any way from contract, specification, or other applicable requirements, or contains variations from the standard production units which meet such specifications and conform to approved drawings, shall be classified and disposed of in accordance with the provisions set forth hereinafter. Other applicable requirements include, but are not limited to, the following:

(a) Applicable welding procedures in accordance with MIL-STD-278 approved by the design review agency.
(b) Applicable materials approved by the design review agency.
(c) Drawings which require design review agency review and those portions of contractor self-approved drawings which reflect specification requirements.

4.1.3.2 Classification. Equipment variations may result from manufacturing errors, choice or quality of materials or inadequate process or procedural controls. Variations involved are those which constitute deviations from specific requirements in the contract or applicable specifications, or which affect installation? maintainability operation, safety of personnel or machinery, quantity or logistics or repair parts, interchangeability of equipment component, subassembly, or any part thereof. Included are defects in materials as determined by tests or revealed by radiography when results of such test or film or both are found to be in excess of limits permitted by applicable standards. These variations fall into two categories and are classified as follows:

(a) Type I - Deviations which involve nonconformance of equipment, material, or production process to a specific requirement in the contract or applicable specification.
(b) Type II - Deviations which adversely affect the above parameters but are within the specific requirements in the contract or specification.

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4.1.3.3 Disposition. Disposition of the equipment variations shall be as follows:

(a) Substandard equipment. Type I equipment variations which render the affected part or parts not suitable for the Navy's intended use by reason of not meeting minimum requirements or standards shall be rejected, in accordance with terms of the contract, except on a case-basis when the design review agency elects to accept such part or parts as suitably modified nonstandard equipment described hereinafter.

(b) Nonstandard equipment. When equipment with variations is determined to be suitable for the Navy's intended use either in the "as-is" or modified condition, such part or parts may be accepted as nonstandard parts in accordance with the procedure set forth hereinafter.

4.1.3.4 Conditions for acceptance. Conditions for acceptance of parts having varia- . tions shall be as follows:

(a) Type I - These variations require contractual action and will be accepted solely at the discretion of the design review agency on a case basis.

(b) Type II - These variations will be approved when all of the following conditions are met:

(1) Effect of the variation(s) is acceptable to the design review agency.

(2) Rate of incidence of such variations is not unreasonable. Where the nature of the variation(s) , the quantity of similar variations, or repetition of the same variations are such as to indicate, in the judgment of the design review agency, lack of proper or effective quality control, or workmanship such variation(s) may not be-approved.

(3) Where parts involved are normally onboard repair parts, 3 of each such nonstandard parts or sets of parts shall be furnished at supplier's expense, consisting of 1 of each as onboard repair parts and 2 of each as stock repair parts.

(4) Where parts involved are normally stock repair parts but not onboard " repair parts, 2 of each such special parts or sets of parts shall be furnished by the suppliers as stock rapair parts.

(S) The repair parts furnished in accordance with (b)(3) and (4) shall consist of the lowest echelon of the parts required to compensate for the variation. For example, the major parts in which the basic error is made need not necessarily be furnished; if a rotor bearing journal is machined undersize necessitating the installation of a nonstandard bearing, repair parts nonstandard bearing shall be furnished, but a repair parts nonstandard rotor need not be furnished.

(6) Separate provisioning documentation for nonstandard repair parts shall be submitted in accordance with MIL-P-15137 (see 6.2). The component involved shall be identified by ship, unit, and component serial number in the documentation.

(7) An equipment variation summary drawing together with certification that resultant special parts have been furnished, shall be submitted for all all type XI variations for each component design furnished under a contract.

4.1.3.5 Procedure for approval of variations. The following procedure shall be used for approval of variations:

(a) Type I and II variations shall be referred to the design review agency for approval via the local DCAS and the procuring agency within 30 days after occurrence. Submittals shall be in the same manner as established for submitting drawings.

(b) Each variation referred to the design review agency shall be accompanied by a letter report containing:

(1) The contract number and item involved.

(2) The date on which the variation was discovered.

(3) The circumstances or conditions under which the variation occurred.

(4) A complete description (including sketch or drawing) of the nature and extent of the variation.

(5) The corrective measures proposed:
a. For the pert or parts containing variations.
b. To prevent the reassurance of similar variations.

(6) The effect of the variation as corrected on performance, endurance, stress levels, and parameters cited in 4.1.3.2 including:
a. Detailed l ngineering basis for acceptance.
b. Certification of the part or parts as corrected.

(7) Identification of the benefit to the Government if variation is accepted as modified.

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4.1.3.6 Any work done on equipment with variations prior to obtaining design review agency approval shall be at the supplier's risk.

4.1.3.7 Copies of all correspondence involving repair parts shall be forwarded to Ships Parts Control Center, Mechanicsburg, Pa.

4.1.3.8 Workmanship and quality control records. Machining errors and equipment variations which are not classified as type I or II (such as nonsignificant and noncritical deviations from drawing dimensions and tolerances or both for castings, forgings, weldments, connections, or machine-processed parts) do not require classification or approval by either the design review agency or the DCAS; however, each such deviation shall be documented, and a copy shall be furnished to the DCAS. If the DCAS considers that an unclassified deviation does involve contractual requirements, then such deviation shall be classified and treated in accordance with the foregoing paragraphs. When a repair restores a part to original drawing dimensions, but involves no change of materials and uses a previously approved repair process, the DCAS shall approve such repair. There shall be periodic reviews between the manufacturer and the DCAS on the accumulated incidents of machining errors for determination of. corrective measures to improve workmanship. The frequency and timing of such meetings shall be as determined by the DCAS.

4.1.4 When specified (see 6.1), intended use makes it mandatory that the pump shall be free of all mercury contamination. During the manufacturing processes, tests and examinations the product to be offered for acceptance shall not come in direct contact with mercury or any of its compounds nor with any mercury containing device employing a single boundary of containment. The manufacturer shall certify that the product, when shipped, is free from mercury contamination.

4.2 Qualification tests.[1/] Tests shall be performed at a laboratory satisfactory to NAVSEC. The several qualification tests on one pump design may be conducted concurrently if practicable. The tests shall fully establish that the product has the reliability and performance capability, specified herein. The following tests shall be performed:

(a) The performance evaluations tests specified in 4.3.2 through 4.3.2.3.
(b) The design evaluation tests specified in 4.3.3 through 4.3.3.3.

4.2.1 The qualification specified herein shall apply separately to one pump of each type. Qualification will include all sizes for a specific type.

4.2.2 The potential supplier desiring qualification of a pump type shall submit design data and calculations, as specified in 3.11 through 3.11.4, to NAVSEC for review and design approval as a prerequisite to the performance of the qualification tests specified herein. The design record, as specified in 3.12 through 3.12.2 shall be submitted upon completion of the qualification tests.

4.3 Tests. The procuring agency and Government representatives shall have the right to examine-test facilities at the suppliers' plants and at the suppliers' subcontractors' plants, and to witness all tests specified herein. All failures, deficiencies and discrepancies revealed during the-performance of the specified tests and the corrective measures taken shall be recorded and fully documented in the applicable test records and test reports. After correction of deficiencies tests shall be repeated to the extent necessary to ascertain acceptability of the modified pump. Major failures indicative of the design deficiency (as distinguished from shop error or faulty workmanship) shall be reported to the design review agency before l correction is made.

4.3.1 Shop tests. Shop tests shall be performed on each pump being supplied (and subcomponents of each pump if applicable). Each shop test shall be fully documented by a prescribed test procedure and a written record of the performance and findings of each test. A compilation of the documentation of all the shop tests performed on each pump, shall be furnished in duplicate to the cognizant Government quality assurance office and to the design review agency. Except when specifically required herein to be tested with the actual driver, shop tests may be conducted using a substitute driver, provided the actual driver separately passes the tests required by its applicable specification. The following shop tests shall be performed on each pump:

(a) Hydrostatic pressure test (see 4.3.1.1).
(b) Over speed test (see 4.3.1.2).

[1/] Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.5 and 6.5.1).

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(c) Mechanical soundness and capacity test (see 4.3.1.3).
(d) Noise test (to be performed only when specified (see 4.3.1.4 and 6.1).
(e) Radiographic examination (see 4.3.1.5).
(f) Thrust test (if specified in 6.1) (see 4.3.2.3).

4.3.1.1 Hydrostatic pressure test. Each pump casing shall be tested hydrostatically to a pressure one and one-half times maximum discharge pressure. The hydrostatic test pressure shall be maintained for at least 30 minutes or longer as necessary for examination of entire casing.

Acceptance criteria: The pump shall exhibit no leakage through the casing material or joints.

4.3.1.2 Over speed test. Each pump driven by a variable speed driver capable of over-speeding shall be operated continuously for 30 minuets at a speed 25 percent above maximum design operating speed. The pump need not be under load except as necessary to prevent damage or injury. Continuous shaft and rigid coupled pumps shall be tested with the driver. Units separately driven through flexible couplings and geared units may be tested by separate pump, gear unit and driver over speed tests. For pumps not requiring a noise test, the over speed test may be conducted on each pump rotating assembly in any casing of the design supplied on the contract or order. The dynamic balance shall be checked by the use of a vibration indicator of a type whereby vibration of an amplitude of 0.001 inch may be readily observed on a scale.

Acceptance criteria: Neither the pump nor the driver shall exhibit noises or roughness of operation. The maximum acceptable readings of the vibration indicator shall not exceed the limits shown in MIL-STD-167 for this equipment.

4.3.1,3 Mechanical soundness and capacity test. This shop test shall be conducted, recorded and reported in accordance with the centrifugal pump Rating Standard and Test Code of the Hydraulic Institute, to the extent that these standards are applicable and are not in conflict with the requirements specified herein. The test record for each pump shall identify the pump by manufacturers' serial number. It shall indicate the diameter of the pump impeller(s) being tested in the pump, it shall include a description (by legend and sketch) of the test stand arrangement including an identifying list of test instruments used and the date when last calibrated, and it shall include the data sheets of all recorded data and sample calculations of data conversion to specified conditions. Test data used for plotting the head-capacity curve shall be corrected to the specified operating conditions. The test shall be performed in three phases, as follows:

(a) Operate the pump and its driver if motor driven, continuously at maximum rated speed and capacity, with the pumped fluid within plus or minus 20°F. of the specified operating temperature (see 6.1) until bearing temperatures (not including water-lubricated bearings) stabilize as manifested by at least three consecutive equal bearing temperature indications taken at intervals of not less than 15 minutes. The pump operation shall be fully monitored throughout this test for proper functioning of packings and seals and bearing lubrication, and for smooth running. The amount of packing leakage shall be measured at regular intervals and recorded. Rolling contact bearing temperature shall meet the requirements of 3.3.9.4.

Acceptance criteria: Operation shall be without over heating and shall be free of abnormal vibrations and noises. Oil temperature rise in force-feed lubricated bearings shall not exceed 50°F. in any bearing and shall not exceed a maximum temperature of 180°F. with inlet cooling water to the oil cooler at 85°F. There shall be no abnormal leakage of water or oil.

(b) Operate the pump at the maximum rated speed, from recirculation flow to as close to free delivery as practicable and with the specified suction conditions prevailing. The unit shall be operated at 7 or more test points, throughout the full operating range, to establish accurately the head-capacity curve at maximum rated speed. The unit shall be operated at each test point until the test values being measured stabilize.

Acceptance criteria: The pump shall deliver the rated capacity and head. The head-capacity curve at maximum rated speed shall satisfy the specified requirements. The total head at all capacities on the curve other than rated capacity shall not deviate by more than plus 5 percent os minus 2 percent of rated heads under 500 feet, plus 3 percent or minus 1 percent of rated heads over 500 feet from the corresponding head on the head-capacity curve at maximum rated speed established in the performance test specified in 4.3.2.1. Bearing temperatures shall have remained normal and within the limits specified.

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(c) Operate the electric motor-driven unit for a minimum of 1 minute in reverse rotation at maximum rated speed.

Acceptance criteria: The unit shall not be damaged by the reverse rotation test. Conformance to hydraulic and noise requirements shall be demonstrated subsequent to the reverse rotation test.

4.3.1.4 Noise test. When required (see 6.1), a noise test shall be conducted and reported in accordance with MIL-STD-740. The noise test stand details, instrumentation, and testing technique shall be submitted by the supplier for review by the design review agency prior to test commencement. Airborne noise tests shall be conducted on the lead unit only on each contract or order.

Acceptance criteria: The unit shall meet the noise level limits specified (see 6.1).

4.3.1.5 Radiographic inspection. Radiographic, magnetic particle and liquid penetrant inspection of castings and weldments shall comply with MIL-STD-278 and MIL-STD-271. Pump casings shall be radiographed in those areas which act as pressure boundaries subject to pressures of 550 pounds per square inch gage (psig) and above.

4.3.2 Performance evaluation tests. One pump of each type, design and size on each contract or order, complete with driver and all appurtenances and controls, shall successfully undergo the performance evaluation tests to establish the complete pump performance map and to ascertain compliance with the specified performance requirements. when pumps of identical type, design and size are being produced under two or more contracts within a 2 year period, only one series of performance evaluation tests shall be required for that production run. The following tests shall comprise the performance evaluatation tests.

(a) Performance test (see 4.3.2.1).
(b) Mechanical shaft seal and alignment procedure (see 4.3.2.2).
(c) Thrust test (if specified in 6.1) (see 4.3.2.3).

4.3.2.1 Performance test. The performance test shall be conducted, recorded and reported in accordance with the requirements for the shop test specified in 4.3.1.3, except that the use of a substitute driver shall not be permitted and that, in addition to the head-capacity curve at maximum rated speed, a full performance map shall be established. The proposed test procedure for this test shall be submitted to the design review agency, for approval approximately at the time of drawing submittal. Test data shall be converted to specified operating conditions for plotting of all performance curves. The performance curves shall be determined at maximum and minimum operating speed for variable speed pumps and at each operating speed for multi-speed pumps. The following curves shall be established;

(a) Capacity versus total head.
(b) Capacity versus pump efficiency.
(c) Capacity versus brake horsepower.
(d) Capacity versus steam consumption or electric power input.-
(e) Capacity versus net positive suction head.

Acceptance criteria: The acceptance criteria specified in 4.3.1.3 (a) and (b) shall be met. The performance map shall exhibit the specified pump performance characteristics and meet the steam rate requirements where applicable. All controls and safety devices shall function as intended throughout the full operating ranges of capacity and speed.

4.3.2.2 Mechanical shaft seal and alignment procedure test. The mechanical seal shall be installed, and alignment done precisely in accordance with the procedures to be used in the technical manual. The mechanical seals shall be run in and the pump operated at maximum rated speed with the fluid pumped at maximum operating temperature. The tests shall verify seal installation and back-up packing installation procedures. Stuffing box leakage, if any, shall be determined. The dynamic balance shall be checked by use of a vibration indicator (see 4.3.1.2).

Acceptance criteria: Mechanical seal leakage shall not exceed 5 drops per minute. The vibrations shall not exceed the limits specified in 4.3.1.2.

4.3.2.3 Thrust test. When required (see 6.1), the pump and driver shall be operated in accordance with the requirements for the shop test specified in 4.3.1.3 at the maximum operating speed. The magnitude and direction of axial thrust loading at the thrust bearing shall be measured. This test may be conducted concurrent with the performance test specified in 4.3.2.1. A curve of thrust versus capacity shall be l stablished. The effect on axial thrust of operating conditions, such as temperature and suction pressure, shall be determined

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by calculation or test for the full range of specified operating conditions, and the results shown on the plot of thrust versus capacity. The effect of rotor position on axial thrust shall be determined by test on the first pump of a design tested, and the results shown on the plot of thrust versus capacity.

Acceptance criteria: The thrust bearing axial load for any specified operating condition or possible rotor position shall not exceed that load used as the basis for design of the thrust bearing.

4.3.2.4 Test reports for the performance evaluation test shall be compiled into a single document. The document shall identify the pump by manufacturer's serial numbers of pump and driver, the contract or order number under which the tests were performed, the ship or class of ships for which the pump is intended if known, and the letter and date of test procedure approval. The test compilation document shall be distributed as follows:

(a) Two to each procuring agency.
(b) Two to one of the following, as applicable.
Defense Contract Administration Services Region (DCASR) .
Defense Contract Administration Services District (DCASD).
Defense Contract Administration Services Office (DCASO).
(c) Two to design review agency.

4.3.3 Design evaluation tests. One pump of each type, design and size complete with the driver and all appurtenance and controls shall successfully undergo the design evaluation test specified herein. Separate design evaluation tests of the pump proper and the driver proper are not acceptable. Design changes which in the opinion of the design review agency may detrimentally affect the reliability of a previously tested and accepted pump design shall be cause to require new design evaluation tests in part or in full. It shall be at the discretion of the design review agency to require new design evaluation tests when an accumulation of several design changes, each of itself apparently not significant to pump reliability, collectively considered render suspect the validity of any one or all of the previously performed design evaluation tests. An endurance test need not be performed provided the vendor can demonstrate to the satisfaction of the design review agency concerned that the pump to be offered has proved satisfactory in previous Navy shipboard service under conditions equally as strenuous as those specified in the ordering data. The proposed design evaluation test procedures shall be submitted to the design review agency for anproval prior to performing the tests. The design evaluation tests shall consist of the following:

(a) Endurance test (see 4.3.3.1).
(b) Shock test (see 4.3.3.2).
(c) Inclined operation test (see 4.3.3.3).
(d) Vibration test (if specified in 6.1) (see 4.3.3.4).

4.3.3.1 Endurance test. The pump shall be operated for a period of not less than 500 hours of actual running time with a minimum of 100 starts to ascertain reliability of performance and operation.

4.3.3.1.1 Prior to commencement of the endurance test and immediately after completion of the 500 hour operating run, the pump and driver shall be disassembled to the extent necessary and the critical dimensions and running clearance, of parts subject to wear, erosion and derangement, measured, calculated and recorded. All components such as turbine rotors and nozzles and pump impellers and casings subject to erosion, corrosion, cavitation, steam . cutting and wear, the effects of which are not subject to routine measurement shall be listed in the inspection record and after completion of the test the condition of each component shall be determined and recorded.

4.3.3.1.2 During the initial and final hours of the endurance test run, capacity and noise tests in l ccordance with 4.3.1.3 and 4.3.1.4 shall be performed to determine the changes in the pump's performance characteristics and noise signature. Vibration measurements shall be taken at the bearin saps or housings of the pump and driver at equal speeds during the initial and final capacity test to determine the changes in mechanical operation.

4.3.3.1.3 The endurance test shall not be continuous but shall be interrupted by at least three rest periods of approximately 8 hours each. A minimum of 109 starts (at full line voltage for motor driven pumps) shall be performed during the course of the test. During an early part of the endurance test the pump shall be operated continuously for 24 hours at a capacity as near free delivery as possible at maximum rated speed, and normal specified temperature, submergence and suction conditions. During the latter part of the endurance test the pump shall be operated as near shut off as possible for 12 hours continuously. The remainder of the endurance test shall be run at maximum rated speed and within

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plus **20°F,** minus 0°F of maximum specified liquid temperature. The pump shall be operated at one third, two thirds and rated capacity in approximately equal time intervals. Operations at rated capacity shall be at minimum specified net positive suction head available, or maximum specified suction lift or vacuum, as applicable.

9 4.3.3.1.4 The pump shall be monitored during the endurance test to accurately record the conditions of operation, the capacity delivered, the total head developed, the speed at which operated and the general performance observed. Data shall be collected and the pump inspected at least twice per day of operation. For each periodic inspection, in addition to all measured data, the record shall indicate the following:

(a) The pump seal or packing leakage rate.
(b) The temperature of the grease and oil-lubricated bearings, ambient air temperature, and pumped fluid temperature.
(c) The airborne noise level (normal-abnormal).
(d) The vibration level (normal-abnormal).
(e) The smoothness of operation (normal-abnormal).
(f) All other abnormal findings.
(g) All adjustments made.
(h) Changes made in the conditions or method of operation.

4.3.3.1.5 Endurance test acceptance criteria:

(a) The head-capacity curve at maximum rated speed after 500 hours of pump operation shall conform to the specification requirements and shall show no abnormal deviations from the curve before the 500 hour test.
(b) The unit performance and operation after 500 hours of operation shall be unchanged and normal and meet all specification requirements.
(c) The unit operation at the end of the l ndurance test shall be smooth and shall exhibit noise and vibration levels that are normal and in conformance with the specification.
(d) Lubrication shall have remained satisfactory throughout the test period. Bearing temperatures shall have remained normal and within the limits specified and shall be consistent with their respective bearing clearances and oil and grease limitations.
The pump, mechanical seal or packing shall have performed as specified. The leakage rate shall not exceed 5 drops per minute for mechanical seals or that recommended by the supplier for packed pumps throughout the test.
(f) All running clearances shall be normal.
(g) All components subject to attack from corrosion, erosion, cavitation, etc. shall be in an acceptable condition commensurate with 500 hours of service.
(h) Wear rates for wearing parts, critical for proper operation shall show a rate of wear for the test period that shall be consistent with the specified design life requirements.
(i) No failures shall have occurred throughout the 500 hour test.

4.3.3.1.6 Post endurance test **procedure.** The unit subjected to the 500 hour endurance teat shall be restored to the as-new conditon by replacement of all parts worn beyond the as-new design tolerances. The restored unit shall successfully pass the shop test; specified in 4.3.1.1, 4.3.1.3(a) end 4.3.1.4 if applicable. The shop test documentation and completed part examination check list shall indicate that the unit was subjected to the l ndurance test and subsequently restored and shop tested and that it shall be certified as fully conforming to the specification for unrestricted service.

4.3.3.2 Shock test. The pump shall undergo a shock test to ascertain that the pump has the necessary shock resistance. The shock test shall be performed in accordance with MIL-S-901 and the specific shock test requirements (see 6.1).

4.3.3.2.1 Before and after the shock test the pump and driver and other components susceptible to internal derangement shall be disassembled to the l xtent necessary and the critical dimensions and running clearances measured, calculated, end recorded. During this disassembly the critical components and assemblies subject to shock damage and derangements shall be identified and listed in the inspection record end after completion of the test the condition of l ach component and assembly determined and recorded. Shafts, impellers and turbine rotors shall be inspected by one of the l pplicable non-destructive test procedures, other than radiography specified in MIL-STD-271.

4.3.3.2.2 Before end after the shock test, shop tests in accordance with 4.3.1.3 shall be performed to determine the changes in performance characteristics of the pump. Vibration measurements shall be taken at the bearing caps or housings of the pump and driver at equal speeds during the initial and final capacity test to determine the changes in mechanical operation.

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4.3.3 .2.3 The unit shall be, mounted on the shock machine or barge" essentially identical to the actual shipboard installation. The procuring agency will furnish the supplier a drawing of the shipboard mounting arrangement. Horizontal pumps when tested in the inclined position on the medium weight shock machine shall be oriented so that the direction of shock is perpendicular to the axis of pump rotation. The pump shall be in operation during the first and third blow in each orientation of the medium weight shock machine. The pumps shall be in operation during the first, third and fifth blows of the shock test on the floating barge. Variable speed pumps shall be operated at the minimum speed and pressure required to insure lubrication of bearings and wearing pasts. Other pumps shall be operated at the lowest rated speed.

4.3.3.2.4 The pump shall be carefully observed during each shock blow and thoroughly examined visually after each blow. After each blow the unit shall be operated at as close to maximum rated speed as possible and checked for abnormal noises and vibrations and proper functioning of controls. Turbine driven pumps may be air driven.

4.3.3.2.5 Shock test acceptance criteria:

(a) There shall be no breakage of parts including mounting bolts.
(b) There shall be no distortion or derangement of any part which would render the unit incapable of performing as specified.
(c) The amplitude of vibration after test at maximum rated speed shall be less than twice the amplitude measured at the same speed before the test.
(d) Lubrication to all bearings shall be maintained.
(e) Critical dimensions and running clearances shall have been maintained.
(f) There shall be no significant change in the head-capacity curve.

4.3.3.2.6 Post shock test procedure. The shock tested unit, if it is to be supplied under a contract or order and if approved shall be restored to the as-new condition by replacement of all parts damaged or-distorted beyond the as-new design tolerances. All rolling contact bearings shall be replaced regardless of condition. The shock tested rolling contact bearings shall be destroyed. The restored unit shall successfully pass the shop tests specified in 4.3.1.1, 4.2.1.3(a). and 4.3.1.4 if applicable. The shop test documentation shall indicate that the unit was subjected to the shock test and subsequently restored, shop tested and examined. The completed part examination check list shall be supplied and shall identify the parts which were replaced (such as the bearings) and shall certify that the unit fully conforms to the specifications for unrestricted service.

4.3.3.3 Inclined operation test. Vertical pumps shall be operated for not less than 30 Minutes inclined at an angle from the normal equal to the combination of the maximum permanent list and trim as specified (see 6.1). Horizontal pumps shall be tested in accordance with the inclined operation tests specified for the driving unit. However, the pump and driver shall be tested as a unit. The use of air or low pressure steam to test turbine driven units is not permitted. Operation in the inclined position shall be as close to rated speed and-capacity as practicable. The performance point shall be indicated in the test report. The pumps shall meet all specified performance requirements without damage while operating in the inclined position.

4.3.3.4 Vibration test. When specified (see 6.1) one pump of a quantity of identical pumps being supplied shall successfully undergo a vibration test in accordance with the requirements of MIL-STD-167 and as supplemented in the contract or order. The vibration test need not be repeated on subsequent contracts or orders for pumps of identical design to those previously tested, provided the previous tests included the frequencies specified (see 6.1).

4.3.3.4.1 The unit shall be mounted on typical shipboard foundations during the vibration test or the shipboard mounting arrangement shall be simulated in springmass characteristics except where this mounting arrangement causes the largest test table capacity to be exceeded. Inability to vibration test the unit because of excessive weight or size shall not release the supplier from furnishing equipment which can withstand the specified vibration inputs. Vibration test l eceptanoe criteria shall be in accordance with MIL-STD-167.

4.3.3.5 Design evaluation test documentation. The test reports for the design evaluation tests shall be prepared in accordance with good l ngineering practice. The test arrangement and procedure, the test events, the test instruments used, the measured data; and the effects, results, and observation shall be accurate and complete l nd shall be presented in a professional manner and in a usable durable form. Distribution of the test reports shall be as follows;

(a) Two copies to the design review agency.
(b) Two copies to the procuring l gency.
(c) Two to one of the following, as applicable:
Defense Contract Administration Services Region (DCASR).,
Defense Contract Administration Services District (DCASD).
Defense Contract Administration Services Office (DCASO).

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4.4 Inspection of preparation for delivery. The packaging, packing and marking shall be inspected for compliance with section 5 of this document.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.6.)

5.1 Preservation, **packaging, packing and marking.** The pump units, accessories, manuals and onboard **repair parts shall b**e preserved, pac age , packed and marked in accordance with MIL-P-16789. The requirements and levels shall be as specified (see 6.1).

6. NOTES "

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type required (see 1.2).
(c) General design requirements (see 3.2).
  (1) Space and weight limitations (see 3.2).
  (2) Hatch diameter (see 3.3.6.4).
  (3) List, pitch, roll, trim, and other ship attitude conditions (see 3.2 and 4.3.3.3).
  (4) Type of mounts (see 3.2.3.2).
  (5) Environmental vibration frequencies (see 3.2.3.3 and 4.3.3.4).
  (6) Attached piping loads (see 3.2.4.1).
  (7) Airborne, structureborne, and fluidborne noise requirements (see 3.2.5.1 and 4.3.1.4).
  (8) Ambient temperature, normal, maximum and minimum.
(d) Performance characteristics (see 3.3.1).
  (1) Fluid to be pumped and specific gravity of fluid.
  (2) Temperature of fluid, normal, maximum, and minimum.
  (3) Raked capacity and range of capacity.
  (4) Rated total head at rated capacity.
  (5) Shutoff head limitation, if any.
  (6) Casing design, test pressures and material.
  (7) Suction head available, normal, maximum and minimum.
  (8) Low suction pressure trip setting range, psi (see 3.7.1 and 3.7.5).
  (9) Limiting rotative speed (see 3.3.1.5).
  (10) Pump arrangement and operation in system, including parallel operation l nd method of control.
  (11) Special performance requirements, if any.
(e) Percent capacity at a pressure not less than 80 percent of rated discharge pressure (see 3.3.1.3).
(f) If externally supplied cooling liquid or externally induced forced draft is required (see 3.3.9.4).
(g) Type of coupling (see 3.3.10.1 and 3.3.10.2).
(h) Connections and fittings.
  (1) Valves, flanges, fittings and bolting required (see 3.4.1.).
  (2) Suction end discharge connections (see 3.3.3).
  (3) Vent and drain valves (see 3.3.2.3).
  (4) Pressure gages required (see 3.4.71.
  (5) Stuffing box gage connection (see 3.3,7.2).
(i) Type of turbine and turbine characteristics (for turbine drive) (see 3.7.1).
  (1) Applicable specification.
  (2) Type of connection to driven unit.
  (3) Steam conditions (pressure and temperature) at the inlet to the trip valve, end maximum boiler steam pressure of the installation for which the turbine is intended (see the specification for pump pressure regulating governors regarding pressure drop through the governor).
  (4) Exhaust conditions (pressure or vacuum) at turbine exhaust.
  (5) Steam consumption limitation, if any.
(j) Type and characteristics of pressure regulating governor (see 3.7.2).
(k) Electrical characteristic (for motor drive) as required by the applicable motor and controller specifications (see 3.7.4 and 3.7.5).
  (1) Applicable specification.
  (2) Voltage, phase characteristics and frequency.
  (3) Service class.
  (4) Ambient temperature.

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(5) Duty.
(6) Enclosure.
(7) Winding (d.c.) or design of motors (a.c.).
(8) Bearings.
(9) Insulation.
(10) Power limitation, if any.
(11) Starting current limitation, if any.
(l) Rotor cradle (sea 3.8.2.5).
(m) Number of sets of special tools (see 3.9.2.6).
(n) Drawings required (see 3.9),
(o) Manuals required and whether complete information for electrical equipment is required (see 3.10.1, 3.10.3 and 3.10,6),
(p) Whether design data and calculations and design record are required (see 3.11 and 3.12).
(q) Quality assurance provisions (see 4.1.2).
(r) Mercury prohibition (see 4.1.4).
(s) Thrust test if required (see 4.3.2(c) and 4.3.2.3).
(t) Vibration test if required (see 4.3.3(d) and 4.3.3.4).
(u) Shock test requirements (see 4.3.3.2) (see 6.1 of MIL-S-901).
(v) Preservation, packaging, packing, marking and shipment requirements (see 5.1).
(w) Identification of procuring agency and design review agency (see 6.4).
(x) Ship type and fluid system in which units are to be installed.
(y) Designation of manufacturer (pump or driver) responsible to the purchaser for compatibility of pump and driver and operation of the assembled unit.

6.1.1 The procurement document should specify if approval is required for the following:

(a) Design drawings (see 3.9.1.2).
(b) Preliminary manuals (see 3.10.1).

6.2 Management control system documents. The following management control system documents should be included on DD Form 1660:

(a) MIL-P-15137 (see 3.8.1, 3.8.2, 3.8.4, 3.9.1.3 and 4.1.3.4(b)(6)).
(b) MIL-I-45208 (see 4.1.2).

6.3 The followings table of pump materials shall be used by the manufacturer in selecting component part materials.

Table III - Pump materials.

Components	Material	Specification	ASTM 1/ 2/ publication
Casings			
Split case (type I and II)	Steel casting - 12 percent chromium	MIL-S-16993, class 1	A296-68,CA15 A351-69,CA15
Barrel type (type III 3/) (outer casings and covers)	Forged steel Corrosion-resistant steel forging - 12 percent chromium	MIL-S-890 MIL-S-862	--------- A336-70, grade F6
Barrel type (type IV) corrosion-resistant steel only	Steel casting - 12 percent chromium	MIL-S-16993, class 1	A296-68, CA15 A351-69, CA15
Discharge nozzles (type III)	Corrosion-resistant steel forging - 12 percent chromium	MIL-S-862	A336-70, grade F6
Suction nozzles (type III)	Corrosion-resistant steel forging - 12 percent chromium or steel casting - 12 percent chromium	MIL-S-862 MIL-S-16993, class 1	A336-70, grade F6 A296-68, CA15 A351-69, CA15
Nozzle head (type IV)	Steel casting - 12 percent chromium or steel casting corrosion-resisting 18 percent chromium	MIL-S-16993, class 1 MIL-S-867, class 1	A296-68, CA15 A351-69, CA15 A296-68, CF8

See footnotes at end of table.

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Table III - Pump materials (continued).

Components	Material	Specification	ASTM 1/ 2/ publication
Antifriction metal 4/	Babbitt (tin base)	QQ-T-390, grade 2 or 3	---
Diaphragm (inter stage pieces)	Steel castings - 12 percent chromium or steel casting corrosion-resisting 18 percent chromium	MIL-S-16993, class 1 MIL-S-867, class 1	A296-68, CA15 A351-69, CA15 A296-68, CF8
Diaphragm bushings 5/ Channel ring bushings Throat bushings Casing wearing rings	Steel castings - 12 percent chromium Corrosion-resisting steel 12 percent chromium Bearing bronze Valve bronze	MIL-S-16993, class 1 QQ-S-763, class 410 QQ-S-764 class 416 QQ-C-390 alloy no. D2 or E8 QQ-C-390, alloy no. D4	A296-68, CA15 A351-69, CA15 A276-70, type 410 A582-70, type 416
Diffusers	Steel castings - 12 percent chromium Steel castings - corrosion-resisting 18 percent chromium	MIL-S-16993, class 1 MIL-S-867, class 1	A296-68, CA15 A351-69, CA15 A296-68, CF8
Impellers 5/ 6/	Steel castings - 12 percent chromium Entire impeller shall be hardened to 302 maximum Brinell	MIL-S-16993, class 1	A296-68, CA15 A351-69, CA15
Shafts	Forged steel Corrosion-resisting steel 12 percent chromium Corrosion-resisting steel 18 percent chromium	MIL-S-890, alloy no. 3 QQ-S-763, class 410 QQ-S-764, class 303	 A276-70, type 410 A582-70, type 303
Shaft sleeves Shaft sleeves under packing or seals	Corrosion resisting steel - 12 percent chromium Corrosion resisting steel - 18 percent chromium Sleeve hardness 7/ 450 BHN minimum Sleeve surface finish 16 RHR maximum	QQ-S-763, class 410 or 420 or QQ-S-764, class 416 QQ-S-763, class 304L weld overlay	A276-70, type 410, or 420 A582-70, type 416, A276-70, type 304L
Water lubricated 8/ bearing sleeves	Bearing bronze casting	QQ-C-390, alloy no. E3, E5 or E8	---
Water lubricated bearing journal sleeves	Corrosion-resisting 12 percent chromium	QQ-S-763, class 410 or QQ-S-764, class 416	A276-70, type 410 A582-70, type 416
Casing bolts, 9/ studs and nuts	Forged steel or corrosion-resisting steel or steel bars	MIL-S-890, MIL-B-857 MIL-S-1222 sym B7 and H	A276-70, type 302, 303, 304 A193-70, grade B7

1/ "ASTM" indicates standard specifications of the American Society for Testing and Materials. (Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19203.)

2/ Detail drawings and lists of material shall reference the specifications actually followed in l ach case, and shall include the class, type or grade of material used in each case, as applicable (see 3.4, 3.9.1.2.2 or 3.9.1.2.4).

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3/ Gasket areas and close fit surfaces of the outer casing and covers shall be overlaid with 12 percent chromium alloy steel or other corrosion-resistant steel except that in non- - pressure areas, overlay is not required.

4/ The bearing shall be capable of operation without perfomance degradation if rebabbitted with the alternate grade of babbitt material.

5/ Opposing wearing surfaces shall be of dissimilar material or the difference in material hardness of opposing surfaces shall be not less than 50 Brinnell.

6/ Variations in physical properties and method of heat treatment will be permitted on an approval basis in order to obtain the desired hardness. If a hardness over 248 BHN is desired and achieved, then an elongation of 12 percent is acceptable for material conforming in all other respects to MIL-S-16993.

7/ An overlay, in conformance with the requirements of 3.4, may be used to achieve the required sleeve hardness.

8/ Special high lead bronze or other suitable bearing materials may be used subject to approval by the design review agency.

9/ Bolts, studs and nuts design shall be in accordance with MIL-B-857.

6.4 Definitions (see 6.1). As used herein , "procuring agency" refers to the activity purchasing the equipment covered by this specification. The procuring agency may be the Government or a shipbuilder. The design review agency "refers to the activity responsible for controlling the technical requirements for design and testing of the equipment. The design review l gency is generally a Government Command or agency such as NAVSHIPS, NAVSEC, or an authorized representative. Communication with the design review agency should be handled through the procuring agency.

6.5 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 17381 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification, The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with 'Provisions Governing Qualification SD-6" (ace 6.5.1).

6.5.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.6 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.7 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE WERE MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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6. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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